Your app works fine with 100 users... but what happens at 1 million?

What is Microservices Architecture?

- Application is divided into small independent services
- Each service handles one specific business function
- Services communicate via APIs or message brokers
- Can be developed, deployed, and scaled independently.

Monolith vs Microservices

- Monolith: Single large codebase, tightly coupled
- Microservices: Multiple small services, loosely coupled
- Monolith is easy to start, hard to scale
- Microservices are complex initially, but highly scalable

Why Microservices in 2026?

- Required for scalable and high-traffic systems
- Used by modern companies for fast deployments
- Supports cloud-native & distributed systems
- High demand in backend & system design roles

Real-World Systems Using Microservices

- Companies like Amazon, Netflix, Uber
- Each feature runs as a separate service
- Enables handling of millions of users simultaneously
- Improves fault isolation & scalability

Problems Microservices Solve

- Scaling: Scale only required services
- Flexibility: Use different technologies per service
- Speed: Faster development & deployments
- Fault Isolation: One service fails, others keep running

Challenges of Microservices

- Complex architecture to manage
- Difficult debugging & monitoring
- Requires DevOps & cloud knowledge
- Network communication adds latency

What You Will Build in This Course

- A complete Amazon-scale backend system
- Services like: Auth, Order, Payment, Notification
- Communication using Kafka & gRPC
- Deployment using Docker & Kubernetes

What You Will Learn in This Course

- Beginner → Advanced → Architect-level concepts
- Real-world system design patterns
- Industry tools like Kafka, Redis, Kubernetes
- Best practices + interview preparation

Prerequisites for This Course

- Basic knowledge of JavaScript / TypeScript
- Understanding of Node.js / NestJS basics
- Basic idea of APIs & backend development
- No prior microservices experience required

How to Get the Best Out of This Course